Year 1 AS Pure Maths

Chapter 10

Trigonometric identities and equations

Practice Booklet

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10.1 Angles in all four quadrants

Practice Book, printed page 78–79.

1. Use the unit circle to determine the quadrants where the following trigonometric functions are positive.

a. $\sin\theta$

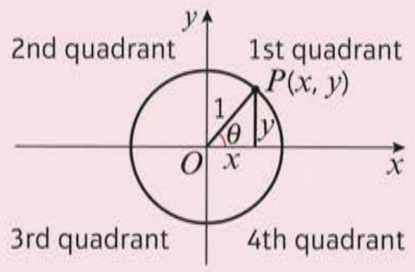
b. $\cos\theta$

c. $\tan\theta$

d. all of $\sin\theta$, $\cos\theta$ and $\tan\theta$

Hint

$P(x, y)$ moves on a unit circle such that OP makes an angle θ with the positive x axis.
$\cos\theta = x = x$-coordinate of P
$\sin\theta = y = y$-coordinate of P
$\tan\theta = \frac{y}{x}$ = gradient of OP
This means that the sign, (+ or −), of the trigonometric functions depends on the sign of x and y.



2. Identify the quadrant that θ lies in as OP moves around the unit circle and hence state whether the trigonometric function of θ is positive or negative.

a. $\theta = 116°$, $\cos\theta$ **b.** $\theta = 240°$, $\tan\theta$ **c.** $\theta = 420°$, $\sin\theta$

d. $\theta = -100°$, $\tan\theta$ **e.** $\theta = 350°$, $\sin\theta$ **f.** $\theta = -80°$, $\cos\theta$

3. Without using a calculator, write down the values of:

a. $\cos 90°$ **b.** $\cos(-180°)$ **c.** $\sin 540°$

d. $\sin(-90°)$ **e.** $\tan 135°$ **f.** $\tan(-45°)$

g. $\sin 225°$ **h.** $\cos(-45°)$ **i.** $\tan(-90°)$

Hint

When OP makes an angle of $45°$ with the positive x-axis the triangle formed is isosceles.
$\cos 45° = x = \frac{1}{\sqrt{2}}$
$\sin 45° = y = \frac{1}{\sqrt{2}}$
$\tan 45° = \frac{y}{x} = 1$

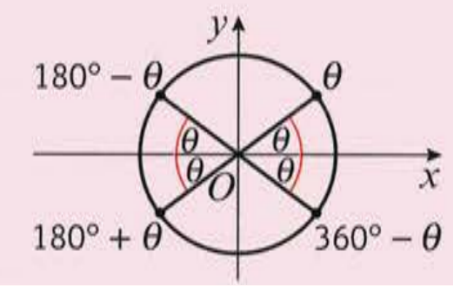
4. Given that θ is an acute angle, express in terms of $\sin\theta$, $\cos\theta$ or $\tan\theta$:

a. $\sin(180° + \theta)$ **b.** $\cos(360° - \theta)$ **c.** $\tan(180° - \theta)$

d. $\sin(-\theta)$ **e.** $\tan(180° + \theta)$ **f.** $\cos(\theta - 180°)$

g. $\sin(\theta - 360°)$ **h.** $\cos(540° - \theta)$ **i.** $\tan(\theta - 180°)$

Hint



5. P Express in terms of trigonometric ratios of acute angles:

a. $\sin 220°$ **b.** $\cos 160°$ **c.** $\tan 210°$

d. $\sin(-100°)$ **e.** $\tan 130°$ **f.** $\cos 280°$

g. $\sin 95°$ **h.** $\cos(-200°)$ **i.** $\tan(-115°)$

10.2 Exact values of trigonometric ratios

Practice Book, printed page 79.

1. Express the following as trigonometric ratios of either 30°, 45° or 60° and hence state the exact value.

a. $\sin 150°$ **b.** $\sin(-45°)$ **c.** $\sin 315°$

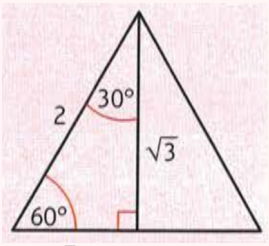
d. $\sin 240°$ **e.** $\cos 150°$ **f.** $\cos(-60°)$

g. $\cos(-135°)$ **h.** $\cos 120°$ **i.** $\tan 225°$

j. $\tan(-150°)$ **k.** $\tan 405°$ **l.** $\tan(-120°)$

Hint

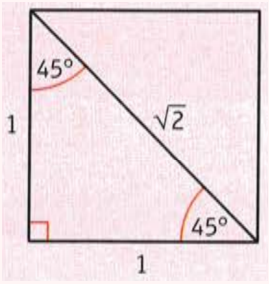
The exact values of the trigonometric ratios of 30° and 60° can be worked out using Pythagoras' theorem by considering an equilateral triangle of side 2 units.



$\sin 30° = \frac{1}{2}$ $\cos 30° = \frac{\sqrt{3}}{2}$ $\tan 30° = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\sin 60° = \frac{\sqrt{3}}{2}$ $\cos 60° = \frac{1}{2}$ $\tan 60° = \sqrt{3}$

The exact values of the trigonometric ratios of 45° can be worked out by considering a square of side 1 unit.



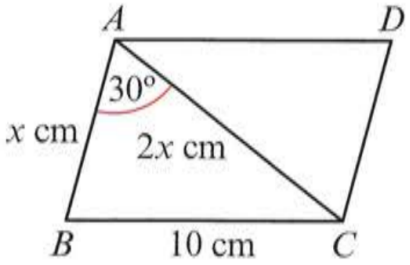
$\sin 45° = \frac{1}{\sqrt{2}}$ $\cos 45° = \frac{1}{\sqrt{2}}$ $\tan 45° = 1$

2. The table shows some of the values of x and y that satisfy the equation $y = a\sin x + b$.

x	0°	30°	60°	90°	120°	150°	180°
y	3	4	$3+\sqrt{3}$	5	$3+\sqrt{3}$	4	3

Find the value of y when $x = 45°$.

3. The diagram shows parallelogram $ABCD$, where $AB = DC = x$ cm and $AD = BC = 10$ cm.



Given that $AC = 2x$ cm and $\angle BAC = 30°$, show that

$$x^2 = \frac{100(5+2\sqrt{3})}{13}$$

10.3 Trigonometric identities

Practice Book, printed page 79–80.

1. Write down the value of:

a. $2\sin^2\theta - (1 - \cos^2\theta) + \cos^2\theta$

b. $6\sin^2\frac{\theta}{2} + 6\cos^2\frac{\theta}{2}$

c. $(1 + \cos x)^2 + (1 - \cos x)^2 + 2\sin^2 x$

Hint

Use the trigonometric identity $\sin^2\theta + \cos^2\theta \equiv 1$.

2. Write in terms of $\tan x$:

a. $\frac{2\sin x}{\cos x}$

b. $\frac{3\sin x}{\sqrt{1-\sin^2 x}}$

c. $\frac{1-\cos^2 x}{1-\sin^2 x}$

Hint

Use the trigonometric identity $\tan x \equiv \frac{\sin x}{\cos x}$ for all values of x such that $\cos x \neq 0$

3. Prove that $\dfrac{\sin^2 x - \cos^2 x}{1 - \sin^2 x} \equiv \tan^2 x - 1$

Hint

To prove an identity, start from the LHS and manipulate the expression until it matches the RHS. You can quote the standard identities in your proof. ← Sections 7.4, 7.5

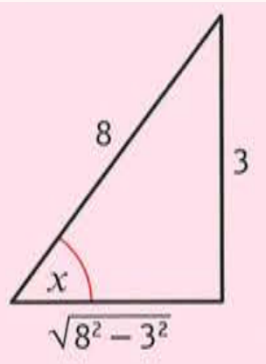
4. Given that $\sin x = \frac{3}{8}$ and that x is obtuse, find the exact value of:

a. $\cos x$

b. $\tan x$

Hint

You can use $\cos^2 x = 1 - \sin^2 x$ to work out the exact value of $\cos x$, then $\tan x = \frac{\sin x}{\cos x}$. Remember you need to check the sign of cosine and tangent in the required quadrant. You can check your answer using basic trigonometry and Pythagoras' theorem in a right-angled triangle.



5. E

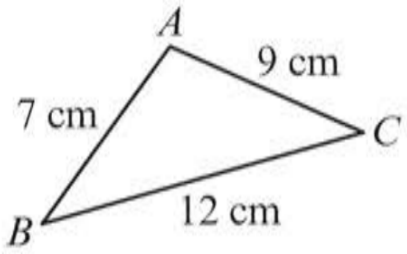
a. Given that $\sin\theta = 6\cos\theta$, find the value of $\tan\theta$. (1 mark)

b. Express $4\cos^2\theta - 5\sin^2\theta$ in terms of $\sin\theta$. (1 mark)

6. E/P Show that the equation $4\cos^2 x - 3\sin^2 x = 2$ can be written as $7\cos^2 x = 5$. (2 marks)

7. E/P Prove that $\dfrac{\cos^2 x - \sin^2 x}{\cos^2 x - \sin x\cos x} \equiv 1 + \tan x$ (3 marks)

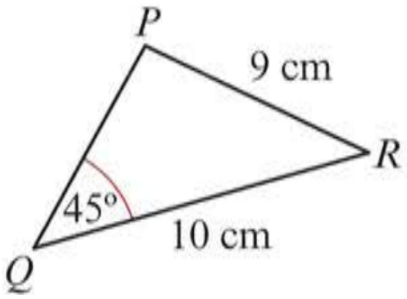
8. E/P The diagram shows the triangle ABC with $AB = 7$ cm, $BC = 12$ cm and $AC = 9$ cm.



a. Find the exact value of $\cos B$, giving your answer in its simplest form. (3 marks)

b. Hence find the exact value of $\sin B$. (2 marks)

9. E/P The diagram shows triangle PQR with $PR = 9$ cm, $QR = 10$ cm and angle $PQR = 45°$.



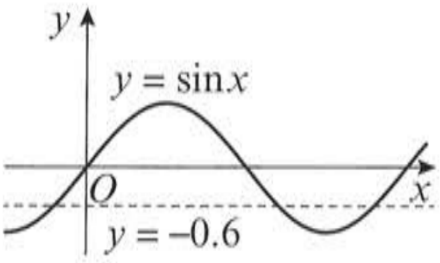
a. Show that $\sin P = \frac{5\sqrt{2}}{9}$ (3 marks)

b. Given that angle P is obtuse, find the exact value of $\cos P$. (2 marks)

10.4 Simple trigonometric equations

Practice Book, printed page 81–82.

1. The diagram shows a sketch of $y = \sin x$.



a. Use your calculator to find the principal solution to the equation $\sin x = -0.6$.

b. Use the graph and your answer to part **a** to find solutions to the equation $\sin x = -0.6$ in the interval $0 \leqslant x \leqslant 360°$.

Hint

Use the inverse trigonometric function on your calculator to get the **principal value**.
Your calculator will give the principal values in these ranges:
$\cos^{-1}$: $0 \leqslant x \leqslant 180°$
$\sin^{-1}$: $-90° \leqslant x \leqslant 90°$
$\tan^{-1}$: $-90° < x < 90°$

2. Find the solutions to the equation $\cos\theta = \frac{1}{\sqrt{2}}$ in the interval $0 \leqslant \theta \leqslant 360°$.

Hint

Find the first solution using your calculator or your knowledge of exact values of trigonometric functions. Draw the graph of $y = \cos\theta$ for the given interval and use the properties of the graph to find the second solution. ← Section 9.5

3. Solve the equation $4\sin x = 3$ in the interval $0 \leqslant x \leqslant 360°$.

Hint

First rewrite the equation in the form $\sin x = \ldots$

4. Solve the equation $\sin\theta = -10\cos\theta$ in the interval $0 < \theta \leqslant 360°$.

Hint

You can use the identity $\tan\theta \equiv \frac{\sin\theta}{\cos\theta}$ to solve trigonometric equations.
The principal value of $\tan\theta$ is not in the given interval. As the period for the graph of $y = \tan\theta$ is 180°, you can find successive solutions by adding or subtracting multiples of 180°.

5. E A student's attempt to solve the equation $5\tan x = 9\sin x$ in the interval $-90° < x < 90°$ is shown.

$$
\begin{aligned}
5\tan x &= 9\sin x \\
5\frac{\sin x}{\cos x} &= 9\sin x \\
5\sin x &= 9\sin x\cos x \\
5 &= 9\cos x \\
\cos x &= \frac{5}{9} \\
x &= 56.3° \text{ (to 3 s.f.)}
\end{aligned}
$$

a. Identify two mistakes made by this student, giving a brief explanation of each mistake. (2 marks)

b. Write down the correct solutions to the original equation. (1 mark)

6. E/P

a. Given that $3\sin\theta = 5\cos\theta$, find the value of $\tan\theta$. (1 mark)

b. Hence, or otherwise, find the values of θ in the interval $0 \leqslant x \leqslant 360°$ for which $3\sin\theta = 5\cos\theta$, giving your answers to 1 decimal place. (3 marks)

7. E/P Find all the values of x for which $\cos^2 x = \frac{3}{4}$ in the interval $-180° \leqslant x < 180°$. (5 marks)

8. E/P

a. Show that the equation $4\cos^2 x - 3\sin^2 x = 1$ can be written as $7\cos^2 x = 4$. (2 marks)

b. Hence solve the equation $4\cos^2 x - 3\sin^2 x = 1$ for $0 \leqslant x \leqslant 360°$, giving your answers to 1 decimal place. (7 marks)

9. E/P

a. Solve $(2 + \tan\theta)(4\sin\theta - 1) = 0$ for $-180° \leqslant \theta < 180°$. (4 marks)

b. Solve $\frac{6\sin x}{\tan x} = 1$ for $0 \leqslant x < 360°$. (6 marks)

10.5 Harder trigonometric equations

Practice Book, printed page 82–83.

1. Find the values of θ in the interval $0 \leqslant \theta \leqslant 360°$ for which:

a. $\cos 3\theta = \frac{\sqrt{3}}{2}$

b. $\sin \frac{\theta}{2} = \frac{1}{\sqrt{2}}$

c. $\tan 2\theta = -\sqrt{3}$

Hint

In part **a**, let $X = 3\theta$. If $0 \leqslant \theta \leqslant 360°$, then $0 \leqslant 3\theta \leqslant 1080°$.
Use the graph of $y = \cos X$ to solve $\cos X = \frac{\sqrt{3}}{2}$ for $0 \leqslant X \leqslant 1080°$.
There will be six solutions. Divide the solutions for X by 3 to find the solutions for θ.

2. Solve the equations in the given intervals:

a. $\sin(\theta + 30°) = 0.2$, $0 \leqslant \theta \leqslant 360°$

b. $\cos(60° - \theta) = -0.4$, $-180° \leqslant \theta \leqslant 180°$

c. $\tan(\theta - 20°) = 9$, $0 \leqslant \theta \leqslant 360°$

Hint

In part **a**, let $X = \theta + 30°$ and solve $\sin X = 0.2$ for $30° \leqslant X \leqslant 390°$.
Subtract $30°$ from each of the solutions for X to find solutions for θ.

3. Solve the equations in the given intervals:

a. $\cos 3\theta = \sqrt{3} \sin 3\theta$, $-180° \leqslant \theta \leqslant 180°$

b. $2 \sin 4\theta - 5 \cos 4\theta = 0$, $0 \leqslant \theta \leqslant 360°$

c. $5 \sin(\theta + 40°) + 8 \cos(\theta + 40°) = 0$, $-180° \leqslant \theta \leqslant 180°$

4. Solve the equations in the given intervals:

a. $\sin(2x - 30°) = 0.75$, $0 \leqslant x \leqslant 360°$

b. $\cos(3x + 20°) = -0.4$, $0 \leqslant x \leqslant 180°$

c. $\tan(4x - 45°) = 2$, $0 \leqslant x \leqslant 180°$

d. $2 \sin(2x + 10°) = 3 \cos(2x + 10°)$, $-180° \leqslant x \leqslant 180°$

Hint

In part **a**, let $X = 2x - 30°$. The end points in the interval for X are $2 \times 0 - 30° = -30°$ and $2 \times 360° - 30° = 690°$. Find the solutions for X, then use $2x - 30° = X$ for each solution to find the solutions for x.

5. E Solve, for $0 \leqslant x < 360°$,

a. $\cos 3x = -\frac{1}{2}$ (6 marks)

b. $\sin(x - 20°) = -\frac{1}{\sqrt{2}}$ (4 marks)

6. E

a. Solve $5 \sin 2x = 3 \cos 2x$ for $0 \leqslant x \leqslant 360°$, giving your answers to 1 decimal place. (5 marks)

b. Find the solutions of the equation $\sin(3x - 15°) = \frac{1}{\sqrt{2}}$ for which $0 \leqslant x \leqslant 180°$. (6 marks)

7. E

a. Find all values of θ, to 1 decimal place, in the interval $0 \leqslant \theta \leqslant 360°$ for which $5 \sin(\theta + 30°) = 4$. (4 marks)

b. Find all values of θ, to the nearest degree, in the interval $0 \leqslant \theta \leqslant 360°$ for which $\tan^2 \frac{\theta}{2} = 16$. (5 marks)

8. E Solve $\tan(5x - 20°) = \frac{5}{2}$ for $-90° < x < 90°$, giving answers to 1 decimal place. (6 marks)

9. E

a. Show that the equation $\tan 2x = 4\sin 2x$ can be written in the form $(1 - 4\cos 2x)\sin 2x = 0$ (2 marks)

b. Hence solve $\tan 2x = 4\sin 2x$ for $0 \leqslant x \leqslant 180°$, giving your answers to 1 decimal place where appropriate. You must show clearly how you obtained your answers. (5 marks)

10.6 Equations and identities

Practice Book, printed page 83–84.

1. Solve the equations for $0 \leqslant \theta \leqslant 360°$.
Give your answers to 3 significant figures where they are not exact.

a. $9\sin^2\theta = 1$ **b.** $4\cos^2\theta - 1 = 0$ **c.** $5\sin^2\theta - \sin\theta = 0$

d. $\tan^2\theta + \tan\theta - 12 = 0$ **e.** $(3\cos\theta - 2)^2 = 1$ **f.** $\tan^2 3\theta = 4$

Hint

You need to be able to solve quadratic equations in trigonometric functions, which can lead to two sets of solutions. ← Section 2.1

2. Solve the equations for $0 \leqslant \theta \leqslant 360°$.
Give your answers to 3 significant figures where they are not exact.

a. $\cos^2 3\theta = 1$

b. $\tan^2\theta = 5\tan\theta$

c. $\sin^2\theta - 3\sin\theta - 4 = 0$

d. $4\sin(\theta + 30°) = \tan(\theta + 30°)$

Hint

There are no valid solutions to $\sin x = k$ or $\cos x = k$ when $k < -1$ or $k > 1$.

3.

a. Show that the equation $2\cos^2 x - \sin^2 x - 4\cos x + 2 = 0$ can be written as $3\cos^2 x - 4\cos x + 1 = 0$

b. Hence solve the equation $2\cos^2 x - \sin^2 x - 4\cos x + 2 = 0$ for $0 \leqslant x \leqslant 360°$.

Hint

You can use $\sin^2 x = 1 - \cos^2 x$ to convert to a quadratic equation in $\cos x$.

4. Solve the equations for $0 \leqslant \theta \leqslant 360°$.
Give your answers to 1 decimal place where they are not exact.

a. $5\cos^2\theta = 6(1 - \sin\theta)$

b. $8 - 7\cos\theta = 6\sin^2\theta$

c. $3\tan\theta = 2\cos\theta$

Hint

You can use $\cos^2\theta = 1 - \sin^2\theta$ and $\tan\theta = \frac{\sin\theta}{\cos\theta}$ to convert to a quadratic equation.

5. E/P

a. Show that the equation $2\sin^2 x - 3\cos^2 x = 1$ can be written as $5\sin^2 x = 4$. (2 marks)

b. Hence solve the equation $2\sin^2 x - 3\cos^2 x = 1$ for $0 \leqslant x < 360°$, giving your answers to one decimal place. (7 marks)

6. E/P Solve $2\cos^2 x + \sin^2 x = \frac{10}{9}$ for $0 < x < 180°$, giving your answers to one decimal place. (4 marks)

7. E/P Find all the solutions of $2\cos^2 x + 2 = 7\sin x$ for $0 \leqslant x \leqslant 360°$. (6 marks)

8.

a. Show that the equation $\sin\theta\tan\theta = 2\cos\theta + 3$ can be written in the form $3\cos^2\theta + 3\cos\theta - 1 = 0$ (3 marks)

b. Hence solve $\sin\theta\tan\theta = 2\cos\theta + 3$ for $0 \leqslant \theta < 360°$, showing each stage of your working. (5 marks)

9. E/P Solve $15\sin^2 x + 8\cos x - 16 = 0$ for $360° \leqslant x < 540°$. Give your answers to one decimal place. (5 marks)

Problem set A

Bronze

a. Given that $2\sin x = 5\cos x$, write down the value of $\tan x$. (1 mark)

b. Hence solve $2\sin(\theta + 20°) = 5\cos(\theta + 20°)$ for $0 \leqslant \theta < 360°$, giving your answers to one decimal place. (5 marks)

Silver

Solve the equation $\dfrac{\sin 3x + 1}{1 - \sin 3x} = 2$, for $0 \leqslant x \leqslant 180°$.
Give your answers to one decimal place. (6 marks)

Gold

Solve the equation $4\cos(2x - 15°) + 3 = 0$, for $-180° \leqslant x \leqslant 180°$.
Give your answers to one decimal place. (7 marks)

Problem set B

Bronze

a. Show that the equation $\dfrac{3\sin^2 x + \cos^2 x}{\cos^2 x} = 5$ can be written as $\tan^2 x = \frac{4}{3}$ (2 marks)

b. Hence solve the equation $\tan^2 x = \frac{4}{3}$ for $0 \leqslant x < 360°$, giving your answers to one decimal place. (6 marks)

Silver

a. Show that the equation $4\cos^2 x + 9\sin x - 6 = 0$ can be written as $4\sin^2 x - 9\sin x + 2 = 0$. (2 marks)

b. Hence solve $4\cos^2 x + 9\sin x - 6 = 0$ for $0 \leqslant x < 720°$, giving your answers to one decimal place. (6 marks)

Gold

Solve $5(\cos 3x - 1) = 3\sin 3x \tan 3x$, for $0 \leqslant x < 90°$, giving your answers, where appropriate, to one decimal place. (9 marks)

Selected exam question bank questions

Question 7

Show that $\dfrac{2\sin^2 x - 1}{\cos^2 x} \equiv \tan^2 x - 1$. (3 marks)

Question 47

Solve $6\tan^2 x = 11\tan x + 10$ for $0 \leq x < 360°$, giving your answers to 1 decimal place. You must show each step of your working. (6 marks)

Question 60

a. Show that the equation $\sin^2 x = 8\cos^2 x - 6\cos x$ can be written in the form $(3\cos x - 1)^2 = 2$ (3 marks)

b. Hence solve $\sin^2 x = 8\cos^2 x - 6\cos x$ for $0 \leq x < 360°$, giving your answers to 2 decimal places. (5 marks)

Question 66

Solve, for $0 \leq x \leq 360°$:

a. $\sin(x + 15°) = \frac{\sqrt{3}}{2}$ (4 marks)

b. $\cos 2x = -\frac{1}{\sqrt{2}}$ giving your answers to 1 decimal place. (6 marks)

Question 76

Solve $\cos(3x - 20°) = -0.6$ for $0 \leq x < 180°$, giving your answers to 1 decimal place. You should show each step in your working. (7 marks)

Question 84

$f(x) = 6x^3 + 11x^2 - 3x - 2$

a. Given that $(x + 2)$ is a factor of $f(x)$, factorise $f(x)$ completely. (6 marks)

b. Given that $g(y) = 6\sin^3 y + 11\sin^2 y - 3\sin y - 2$, find the values of y that satisfy $g(y) = 0$ for $0 < \theta \leq 360°$, giving your answers to 1 decimal place where appropriate. (3 marks)

Question 85

a. Solve the equation $5\sin(2\theta + 60°) = 4$ for $0 \leq \theta < 360°$, giving your answers to 1 decimal place. You must show each step of your working. (6 marks)

b. Solve the equation $8\tan x - 3\cos x = 0$ for $360° \leq x \leq 540°$, giving your answers to 2 decimal places. (5 marks)

Answers

10.1 Angles in all four quadrants

1.

a. 1st and 2nd
b. 1st and 4th
c. 1st and 3rd
d. 1st

2.

a. 2nd, negative
b. 3rd, positive
c. 1st, positive
d. 3rd, positive
e. 4th, negative
f. 4th, positive

3.

a. 0
b. -1
c. 0
d. -1
e. -1
f. -1
g. $-1/\sqrt{2}$
h. $1/\sqrt{2}$
i. undefined

4.

a. $-\sin\theta$
b. $\cos\theta$
c. $-\tan\theta$
d. $-\sin\theta$
e. $\tan\theta$
f. $-\cos\theta$
g. $\sin\theta$
h. $-\cos\theta$
i. $\tan\theta$

5.

a. $-\sin 40^\circ$
b. $-\cos 20^\circ$
c. $\tan 30^\circ$
d. $-\sin 80^\circ$
e. $-\tan 50^\circ$
f. $\cos 80^\circ$
g. $\sin 85^\circ$
h. $-\cos 20^\circ$
i. $\tan 65^\circ$

10.2 Exact values of trigonometric ratios

1.

a. $\sin 30^\circ = \frac{1}{2}$

b. $-\sin 45^\circ = -\frac{1}{\sqrt{2}}$

c. $-\sin 45^\circ = -\frac{1}{\sqrt{2}}$

d. $-\sin 60^\circ = -\frac{\sqrt{3}}{2}$

e. $-\cos 30^\circ = -\frac{\sqrt{3}}{2}$

f. $\cos 60^\circ = \frac{1}{2}$

g. $-\cos 45^\circ = -\frac{1}{\sqrt{2}}$

h. $-\cos 60^\circ = -\frac{1}{2}$

i. $\tan 45^\circ = 1$

j. $\tan 30^\circ = \frac{\sqrt{3}}{3}$

k. $\tan 45^\circ = 1$

l. $\tan 60^\circ = \sqrt{3}$

2. $3 + \sqrt{2}$

3. $10^2 = x^2 + (2x)^2 - 2 \times x \times 2x \cos 30^\circ$

$$\Rightarrow 100 = 5x^2 - 4x \times x \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow 100 = x^2(5 - 2\sqrt{3})$$

$$\Rightarrow x^2 = \frac{100}{5 - 2\sqrt{3}} = \frac{100(5 + 2\sqrt{3})}{13}$$

10.3 Trigonometric identities

1.

a. 1

b. 6

c. 4

2.

a. $2\tan x$

b. $3\tan x$

c. $\tan^2 x$

3. $\text{LHS} \equiv \dfrac{\sin^2 x - \cos^2 x}{1 - \sin^2 x} \equiv \dfrac{\sin^2 x - \cos^2 x}{\cos^2 x}$

$$\equiv \frac{\sin^2 x}{\cos^2 x} - \frac{\cos^2 x}{\cos^2 x} \equiv \tan^2 x - 1 \equiv \text{RHS}$$

4.

a. $-\sqrt{55}/8$

b. $-3/\sqrt{55}$

5.

a. 6

b. $(2 - 3\sin\theta)(2 + 3\sin\theta)$ or $4 - 9\sin^2\theta$

6. Using $\sin^2 x \equiv 1 - \cos^2 x$,
$4\cos^2 x - 3(1 - \cos^2 x) = 2 \Rightarrow 7\cos^2 x = 5$

7. $\text{LHS} \equiv \dfrac{\cos^2 x - \sin^2 x}{\cos^2 x - \sin x \cos x} \equiv \dfrac{\cos x + \sin x}{\cos x}$

$$\equiv \frac{\cos x}{\cos x} + \frac{\sin x}{\cos x} \equiv \text{RHS}$$

8.

a. $\frac{2}{3}$

b. $\frac{\sqrt{5}}{3}$

9.

a. Use $\sin 45° = \frac{1}{\sqrt{2}}$ and the sine rule $\frac{\sin P}{10} = \frac{\sin 45°}{9}$

b. $-\frac{\sqrt{31}}{9}$

10.4 Simple trigonometric equations

1.

a. $-36.9°$ (1 d.p.)

b. $x = 216.9°$, $x = 323.1°$ (1 d.p.)

2. $\theta = 45°, 315°$

3. $x = 48.6°, 131.4°$ (1 d.p.)

4. $\theta = 95.7°, 275.7°$ (1 d.p.)

5.

a. Line 3: Divided both sides by $\sin x$ and lost the solution(s) for $\sin x = 0$.
Line 6: Has not identified the negative solution for $\cos x = \frac{5}{9}$.

b. $x = -56.3°, 0.0°, 56.3°$ (1 d.p.)

6.

a. $\frac{5}{3}$

b. $59.0°, 239.0°$

7. $\pm 30°, \pm 150°$

8.

a. Use $\sin^2 x = 1 - \cos^2 x$: $4\cos^2 x - 3(1 - \cos^2 x) = 1 \Rightarrow 4\cos^2 x - 3 + 3\cos^2 x = 1$

b. $x = 40.9°, 139.1°, 220.9°$ or $319.1°$

9.

a. $x = -63.4°, 14.5°, 116.6°, 165.5°$ (1 d.p.)

b. $x = 80.4°$ or $279.6°$ (1 d.p.)

10.5 Harder trigonometric equations

1.

a. $10°, 110°, 130°, 230°, 250°, 350°$

b. $90°, 270°$

c. $60°, 150°, 240°, 330°$

2.

a. $\theta = 138.5°, 341.5°$ (1 d.p.)

b. $\theta = -53.6°, 173.6°$ (1 d.p.)

c. $\theta = 103.7°, 283.7°$ (1 d.p.)

3.

a. $\theta = -170°, -110°, -50°, 10°, 70°, 130°$

b. $\theta = 17.0°, 62.0°, 107.0°, 152.0°, 197.0°, 242.0°, 287.0°, 332.0°$ (1 d.p.)

c. $\theta = -98°, 82°$ (nearest degree)

4.

a. $x = 39.3°, 80.7°, 219.3°, 260.7°$ (1 d.p.)

b. $x = 31.2°, 75.5°, 151.2°$ (1 d.p.)

c. $x = 27.1°, 72.1°, 117.1°, 162.1°$ (1 d.p.)

d. $x = -156.8°, -66.8°, 23.2°, 113.2°$ (1 d.p.)

5.

a. $x = 40°, 80°, 160°, 200°, 280°, 320°$

b. $x = 245°, 335°$

6.

a. $x = 15.5°, 105.5°, 195.5°, 285.5°$

b. $x = 20°, 50°, 140°, 170°$

7.

a. $23.1°, 96.9°$

b. $152°, 208°$

8. $x = -54.4°, -18.4°, 17.6°, 53.6°$ or $89.6°$ (1 d.p.)

9.

a. $\dfrac{\sin 2x}{\cos 2x} = 4\sin 2x \Rightarrow \sin 2x = 4\sin 2x\cos 2x$

$\Rightarrow \sin 2x - 4\sin 2x\cos 2x = 0$

b. $\sin 2x = 0$ gives $x = 0°, 90°, 180°$

$\cos 2x = \frac{1}{4}$ gives $x = 37.8°, 142.2°$

10.6 Equations and identities

1.

a. $\theta = 19.5°, 161°, 199°, 341°$

b. $\theta = 60°, 120°, 240°, 300°$

c. $\theta = 0°, 11.5°, 168°, 180°, 360°$

d. $\theta = 71.6°, 104°, 252°, 284°$

e. $\theta = 0°, 70.5°, 289°, 360°$

f. $\theta = 21.1°, 38.9°, 81.1°, 98.9°, 141°, 159°, 201°, 219°, 261°, 279°, 321°, 339°$

2.

a. $\theta = 0°, 60°, 120°, 180°, 240°, 300°, 360°$

b. $\theta = 0°, 78.7°, 180°, 259°, 360°$

c. $\theta = 270°$

d. $\theta = 45.5°, 150°, 254°, 330°$

3.

a. $2\cos^2 x - (1 - \cos^2 x) - 3\cos x + 2 = 0 \Rightarrow 3\cos^2 x - 4\cos x + 1 = 0$

b. $x = 0°, 70.5°, 289.5°, 360°$

4.

a. $\theta = 11.5°, 90°, 168.5°$

b. $\theta = 48.2°, 60°, 300°, 311.8°$

c. $\theta = 30°$ or $150°$

5.

a. $2\sin^2 x - 3(1 - \sin^2 x) = 1 \Rightarrow 5\sin^2 x = 4$

b. $x = 63.4°, 116.6°, 243.4°, 296.6°$

6. $x = 70.5°, 109.5°$

7. $x = 30°, 150°$

8.

a. $\dfrac{\sin\theta\sin\theta}{\cos\theta} = 2\cos\theta + 3 \Rightarrow \sin^2\theta = 2\cos^2\theta + 3\cos\theta$

$\Rightarrow 3\cos^2\theta + 3\cos\theta - 1 = 0$

b. $74.7°, 285.3°$

9. $\theta = 430.5°$ or $438.5°$

Problem-solving Set A

Bronze

a. $\frac{5}{2}$

b. $\theta = 48.2°, 228.2°$

Silver $x = 6.5°, 53.5°, 126.5°, 173.5°$

Gold $x = -103.2°, -61.8°, 76.8°, 118.2°$

Problem-solving Set B

Bronze

a. $\frac{3\sin^2 x+\cos^2 x}{\cos^2 x} = \frac{3\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x}$
Use $\tan^2 x = \frac{\sin^2 x}{\cos^2 x}$ and rearrange equation.

b. $x = 49.1°$, $130.9°$, $229.1°$, $310.9°$

Silver

a. $4(1 - \sin^2 x) + 9\sin x - 6 = 0 \Rightarrow 4\sin^2 x - 9\sin x + 2 = 0$

b. $x = 14.5°$, $165.5°$, $374.5°$, $525.5°$

Gold $x = 0°$, $37.3°$, $82.7°$

Selected exam question bank answers

Question 7 $$\text{LHS} \equiv \frac{2\sin^2 x - (\sin^2 x + \cos^2 x)}{\cos^2 x} \equiv \frac{\sin^2 x - \cos^2 x}{\cos^2 x}$$
$$\equiv \frac{\sin^2 x}{\cos^2 x} - \frac{\cos^2 x}{\cos^2 x} \equiv \tan^2 x - 1 \equiv \text{RHS}$$

Question 47 $x = 68.2°$, $146.3°$, $248.2°$ or $326.3°$

Question 60

a. Use $\sin^2 x = 1 - \cos^2 x$ and rearrange

b. $x = 36.42°$, $97.94°$, $262.06°$ or $323.58°$

Question 66

a. $x = 45°$ or $105°$

b. $x = 67.5°$, $112.5°$, $247.5°$ or $292.5°$

Question 76 $x = 49.0°$, $84.4°$ or $169.0°$

Question 84

a. $f(x) = (x + 2)(2x - 1)(3x + 1)$

b. $y = 30°$, $150°$, $199.5°$ or $340.5°$

Question 85

a. $\theta = 33.4°$, $176.6°$, $213.4°$ or $356.6°$

b. $x = 379.47°$ or $520.53°$